

To whom it may concern

Re: Recommendation letter for Ms. Yan Yan



Universität Zürich

ETH zürich

Dear Sir or Madam,

Xosé Luís Deán-Ben, PhD

It is my pleasure to recommend Ms. Yan Yan, who is conducting her research project in our Multiscale Functional and Molecular Imaging Lab group at the Institute for Biomedical Engineering, ETH Zurich, under my supervision.

Faculty of Medicine and Institute of Pharmacology and Toxicology, University of Zürich, Switzerland

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During her time in the lab, Yan worked on model-based and deep learning-driven optoacoustic image reconstruction, contributing to the development of a hybrid framework that integrates physics-based modeling with neural network-based regularization. Her work required bridging theoretical understanding and practical implementation, and she handled both aspects with impressive maturity.

Yan demonstrated strong analytical thinking, programming ability, and technical adaptability. She mastered the theoretical background of inverse problems, implemented iterative reconstruction algorithms in MATLAB, and explored how to combine these methods with convolutional neural networks (CNNs) in Python to improve image quality and reconstruction speed. Her ability to link mathematical modeling with data-driven learning shows a deep understanding of interdisciplinary research.

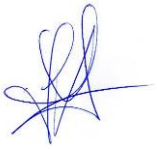
29th October, 2025

Beyond her technical skills, Yan exhibited a reflective and systematic approach to research. She regularly organized and summarized her progress, producing clear workflow diagrams to visualize and optimize the reconstruction pipeline. Her habit of dissecting complex code and reasoning through its underlying logic reflects a genuine curiosity and a commitment to deep conceptual understanding rather than surface-level implementation. Yan's interdisciplinary mindset—actively connecting computational methods, physics, and machine learning—made her a valuable contributor to our research discussions.

In addition to her technical and analytical strengths, Yan is highly motivated, independent, and communicative. She made steady progress, sought constructive feedback, and incorporated it thoughtfully into her work. Her persistence, attention to detail, and intellectual curiosity make her a promising young researcher capable of excelling in a demanding academic environment.

I am confident that Yan possesses the maturity, competence, and creativity required to succeed in any doctoral program. I therefore recommend her without reservation, believing she will make valuable contributions to whichever field of research she chooses to pursue.

Sincerely,



Dr. X. Luís Deán-Ben

Group Leader

Multiscale Functional and Molecular Imaging Lab



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